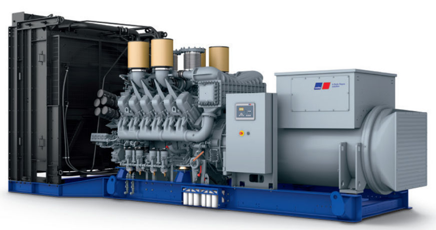
Diesel Generator Set

mtu 12V4000 DS1650

380V – 11 kV/50 Hz/prime power/fuel consumption optimized 12V4000G14F/water charge air cooling



Optional equipment and finishing shown. Standard may vary.

Product highlights

Benefits
- Low fuel consumption
- Optimized system integration ability
- High reliability
- High availability of power
- Long maintenance intervals

Support
- Global product support offered

Standards
- Engine-generator set is designed and manufactured in facilities certified to standards ISO 2008:9001 and ISO 2004:14001
- Generator set complies to ISO 8528
- Generator meets NEMA MG1, BS 5000, ISO, DIN EN and IEC standards
- NFPA 110

Power rating
- System ratings: 1490 kVA - 1600 kVA
- Accepts rated load in one step per NFPA 110*
- Generator set complies to G3 according to ISO 8528-5
- Generator set exceeds load steps according to ISO 8528-5*

Performance assurance certification (PAC)
- Engine-generator set tested to ISO 8528-5 for transient response
- 75% load factor
- Verified product design, quality and performance integrity
- All engine systems are prototype and factory tested

Complete range of accessories available
- Control panel
- Power panel
- Circuit breaker/power distribution
- Fuel system
- Fuel connections with shut-off valve mounted to base frame
- Starting/charging system
- Exhaust system
- Mechanical and electrical driven radiators
- Medium and oversized voltage alternators

Emissions
- Fuel consumption optimized

Certifications
- CE certification option
- Unit certificate acc. to VDE-AR-N 4110

* Changes to the standard parameter sets (alternator-regulator and genset-controller) are necessary

A Rolls-Royce solution

Application data [1)]

Engine

Manufacturer	*mtu*
Model	12V4000G14F
Type	4-cycle
Arrangement	12V
Displacement: l	57.2
Bore: mm	170
Stroke: mm	210
Compression ratio	16.4
Rated speed: rpm	1500
Engine governor	ECU 9
Max power: kWm	1420
Air cleaner	dry

Fuel system

Maximum fuel lift: m	5
Total fuel flow: l/min	16

Fuel consumption [2)]	l/hr	g/kwh
At 100% of power rating:	323.3	189
At 75% of power rating:	250.2	195
At 50% of power rating:	173.7	203

Liquid capacity (lubrication)

Total oil system capacity: l	260
Engine jacket water capacity: l	160
Intercooler coolant capacity: l	40

Combustion air requirements

Combustion air volume: m^3/s	1.6
Max. air intake restriction: mbar	50

Cooling/radiator system

Coolant flow rate (HT circuit): m^3/h	56
Coolant flow rate (LT circuit): m^3/h	30
Heat rejection to coolant: kW	540
Heat radiated to charge air cooling: kW	200
Heat radiated to ambient: kW	75
Fan power for electr. radiator (40°C): kW	38

Exhaust system

Exhaust gas temp. (after turbocharger): °C	460
Exhaust gas volume: m^3/s	4.0
Maximum allowable back pressure: mbar	85
Minimum allowable back pressure: mbar	30

Standard and optional features

System ratings (kW/kVA)

Generator model	Voltage	fuel consumption optimized					
		without radiator			with mechanical radiator		
		kWel	kVA*	AMPS	kWel	kVA*	AMPS
Leroy Somer LSA52.3 S5 (Low voltage Leroy Somer standard)	380 V	1280	1600	2431	1240	1550	2355
	400 V	1280	1600	2309	1240	1550	2237
	415 V	1280	1600	2226	1240	1550	2156
Marathon 743RSL7090 (Low voltage Marathon)	380 V	1272	1590	2416	1232	1540	2340
	400 V	1264	1580	2281	1240	1550	2237
	415 V	1192	1490	2073	1192	1490	2073
Marathon 744RSL7091 (Low voltage Marathon oversized)	380 V	1272	1590	2416	1232	1540	2340
	400 V	1264	1580	2281	1240	1550	2237
	415 V	1192	1490	2073	1192	1490	2073
Marathon 1020FDH7095 (Medium volt. Marathon)	11 kV	1272	1590	83	1232	1540	81
Leroy Somer LSA53.2 VL6 (Med. volt. Leroy Somer)	11 kV	1272	1590	83	1240	1550	81

* cos phi = 0,8

1 All data refers only to the engine and is based on ISO standard conditions (25°C and 100m above sea level).

2 Values referenced are in accordance with ISO 3046-1. Conversion calculated with fuel density of 0.83 g/ml. All fuel consumption values refer to rated engine power.